

DARES_Electronics_Project_Structure

Competition Flight-Control & Electrical System
+ Motor Test Bench
Updated 15 September 2026 for Jinghan Li

Purpose of this revision. This version corrects the earlier six-parallel-project framing. Electronics currently has **two major engineering objectives**. It also includes the project progress since 2025 and revises parts of the guidance. As this is a student-led team, the initial reference materials should be selected by each team leader rather than being fully prescribed in advance. This is intended to give team leaders more autonomy, a stronger sense of ownership, and a greater sense of achievement in leading their own projects.

Organisation	DARES, University of Melbourne
Aircraft	Fixed-wing UAV for NFC 2027
Scope	Avionics, propulsion electronics, controls and test infrastructure
Engineering leadership	Jinghan Li — Engineering Manager , responsible for engineering direction, technical structure and final engineering decisions.
Electrical Leader	Not yet appointed. After Week 10, one Electrical Leader will be selected from the current subsystem/project leaders.
Decision status	Proposed ownership remains subject to confirmation at an officer/management meeting.

Beginner Onboarding & Project Entry

PURPOSE

Give new members a common, safe baseline before they enter a technical project. No previous experience is assumed. Complete the four milestones below in order.

Entry milestones

Complete and review each milestone before moving to the next.

01 Download the Ground Station

Download and install the team-selected ground station. Open it successfully and connect to the approved simulator or training flight controller.

02 Complete Basic Wiring

Complete the approved training wiring for power, flight controller, receiver, telemetry, servos, ESC and required sensors. Check polarity, pinout and signal ground before power-on.

03 Complete Component Selection

Complete a short selection sheet for an assigned subsystem. Check voltage, current, interface or protocol, connector and overall system compatibility.

04 Complete System Setup

Complete basic configuration and calibration. Verify RC mapping, actuator direction, failsafe, telemetry and data logging during a bench check.

Project Team Entry

After all four milestones have been completed and reviewed by a project lead or experienced member, the beginner may join a project team and take a clearly scoped task.

Ground station -> Wiring -> Component selection -> Setup -> Project team

System overview

01 Competition Flight-Control & Electrical System Integrated aircraft system

PURPOSE Develop the complete electronics and control system required for the NFC fixed-wing aircraft.

CURRENT DEVELOPMENT Flight control; motor control / ESC; and battery management / power. Sensors and integration support these three streams; vision recognition is a separate future-facing task.

PRINCIPLE Hardware and software are developed together inside each subsystem. The subsystem boundaries are engineering interfaces, not automatically separate projects.

Control separation Aircraft flight-control law → motor/actuator command → ESC. The ESC currently uses six-step commutation and will later develop FOC. The aircraft flight controller may use a different control method; its control law remains to be selected.

02 Motor Test Bench Independent engineering deliverable

ROLE IN CURRENT PLAN Use the existing test bench to validate the present six-step ESC, support the later FOC implementation, and measure propulsion-system behaviour for the competition aircraft.

MEASUREMENTS Thrust; RPM; voltage; current; electrical power; motor/ESC temperature; efficiency; command input; propeller performance.

EXISTING BASE Preserve and extend the current `Motor_testing_bench` repository and its history.

OWNERSHIP **TBD** Existing contributors and the next project owner must be confirmed.

Structure rule. A new project should require an independent deliverable, defined inputs and outputs, sufficient work and active members, and a clear reason it cannot be managed as a subsystem of an existing project.

High-level team and project structure

The diagram supplements the detailed workstream definitions. The Electrical Leader position remains unfilled; after Week 10, one person will be selected from the confirmed subsystem/project leaders.

Place `team_structure.png` beside this `.tex` file.

Current project workstreams

The current electrical project structure consists of the following three technical projects. Supporting functions and future research are listed separately and are not additional current projects.

1.1 Flight & Attitude Control Proposed: Rohan Lynch + Ash Zhang

PROJECT BASIS Develop the aircraft-level guidance and control system for the NFC fixed-wing aircraft. It converts estimated aircraft states and pilot or mission objectives into actuator and propulsion commands. This is a separate control layer from the ESC motor-control algorithm.

FUNCTIONAL BOUNDARY In: state estimate, pilot or mission setpoints, link and system-health status. Out: aileron, elevator, rudder and throttle commands; flight-mode and failsafe actions; logged controller states for the FDR.

CURRENT STATE ARCHITECTURE AND CONTROL METHOD NOT YET SELECTED The team must first define the aircraft model, controlled states, sensor inputs, actuator outputs, performance targets and safety requirements. A COTS autopilot may be used as the baseline or reference, while selected functions may later be replaced with in-house implementations.

DEVELOPMENT PATH Define requirements and the aircraft model; select a baseline; run fixed-wing SITL and HIL; validate the rate loop, attitude loop and mixer or control allocation; then validate first flight, mission modes and failsafe behaviour. Conventional, testable control should establish the baseline before advanced approaches become competition-critical.

VALIDATION Compare predicted, simulated and measured flight behaviour, and record controller states and flight data so the FDR can be checked against the actual aircraft response.

OWNERSHIP Rohan and Ash are proposed for this workstream. They should select the initial references and implementation approach, subject to confirmed interfaces, safety requirements and validation evidence.

Required interface The flight controller sends actuator and propulsion commands. The ESC interprets the propulsion command and performs lower-level commutation and current control. The two controllers are designed separately but validated together.

1.2 Motor Control & ESC Transition: Jinghan → Jayden + Shangrui

PROJECT BASIS Develop the BLDC motor-drive and ESC hardware and firmware for the propulsion system.

CURRENT METHOD SIX-STEP COMMUTATION The present ESC control implementation uses six-step commutation. Existing assets include the ESC design, basic host-side software, STM32F1/F4-class development hardware purchased personally by Jinghan, and ODrive-related development hardware.

NEXT CONTROL STAGE FOC DEVELOPMENT After the six-step implementation is stable and testable, the next technical stage is to develop field-oriented control. This requires a defined current-sensing and rotor-angle strategy, Clarke/Park transforms, d/q current regulation and a PWM modulation method, followed by bench validation.

OWNERSHIP Jinghan provides technical handover and integration support. Jayden is proposed for ESC and motor-control implementation. Shangrui may contribute to control integration and advanced approaches. Their boundary still needs discussion.

1.3 Battery Management & Power Proposed transition: Zixi → Zi Wang

PROJECT BASIS The BMS is being developed from the open-source [FRBP-ESM-111A-KY reference design on OSHWHub](#). It should be treated as an adaptation and verification project, not a new design starting from zero.

CURRENT TASK Review the reference schematic, PCB, BOM and operating assumptions against the aircraft battery-pack requirements. Record every retained element and every DARES-specific modification.

VALIDATION Verify cell measurement, pack voltage/current limits, temperature monitoring, protection thresholds, balancing behaviour, charge/discharge behaviour, power distribution and ESC/avionics interfaces before aircraft use.

OWNERSHIP Zixi was the previous primary BMS owner but currently has limited availability because of an internship. Zi Wang (Alfred) is a possible successor/supporting owner. The handover and appointment require confirmation.

Next stage Freeze the aircraft pack requirements; complete a design-difference review against FRBP-ESM-111A-KY; define the bench-test plan; then approve or revise the adapted hardware.

Supporting and future work

Sensors & State Estimation Supporting function; owner TBD

PURPOSE Turn noisy and biased sensor data into a state estimate the flight controller can trust.

SCOPE IMU integration; filtering and sensor fusion; attitude/state estimation; Kalman/EKF methods; GPS, barometer, magnetometer, pitot/airspeed and flight-data logging.

STRUCTURE Initially keep this inside the competition flight-control system. It may become a separate workstream later if staffing and deliverables justify it.

Start here Define required states and update rates; choose COTS autopilot versus custom sensor hardware; define ground truth and logging; implement a basic complementary filter before expanding complexity.

Control Integration & Advanced Control Future R&D; proposed: Shangrui Wu

ROLE Act as a technical bridge between motor-level control and aircraft-level attitude/flight control.

FIT Shangrui has previous motor-control experience and current interests in control theory, flight-control laws and reinforcement learning.

GUARDRAIL Advanced control and RL should begin as an R&D stream. They should not replace a validated conventional architecture or become a competition critical-path dependency before evidence supports the change.

Vision Recognition & Higher-Level Autonomy Separate task; vision recognition: David

POTENTIAL SCOPE Vision-based detection; object/target recognition; navigation assistance; autonomous decision-making; higher-level control algorithms.

STATUS **NOT YET A MATURE CORE PROJECT** Begin only if competition requirements, staffing and a concrete deliverable justify it.

OWNERSHIP David is responsible for the vision-recognition work. Any expansion into broader autonomy or aircraft-control responsibilities remains a separate engineering decision under the Engineering Manager.

Proposed ownership map

Area	Previous/current state	Proposed direction
Flight & attitude	Architecture and control method not yet selected	Rohan + Ash; define requirements and baseline, then validate SITL/HIL and selected control methods
Motor control / ESC	Six-step commutation	Stabilise six-step, then develop FOC; Jayden + Shangrui
Battery / BMS / power	Based on FRBP-ESM-111A-KY	Adapt, review and validate; Zi Wang subject to confirmation
Control integration	Previously implicit	Shangrui
Sensors / estimation	Not clearly assigned	TBD
Vision recognition	Future workstream	David
Motor test bench	Existing project and repository / existing contributors	

This table is a proposed responsibility map, not a record of final appointments. Each proposed lead should be consulted before confirmation.